Do these exercises as an assignment on electric field

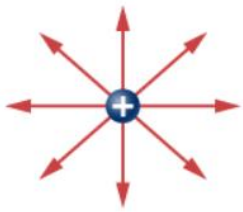
1.Define electric field.

2. Explain the role of a test charge

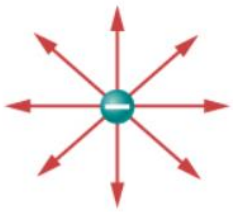
2.State Coulomb's law.

3. Give an example on where electric dipole are found in nature

4. Which of the following electric field lines are incorrect for point charges? Explain why.



(a)



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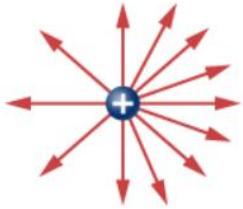

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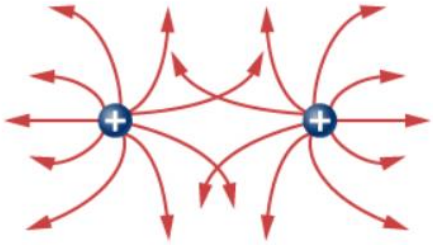

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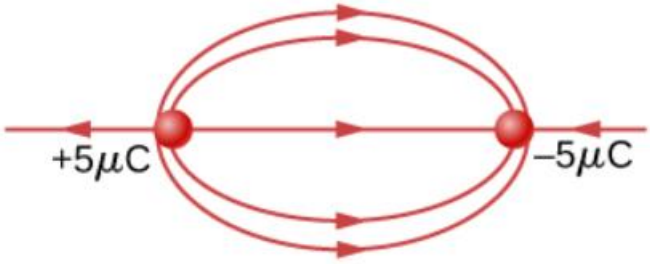


(g)

5. State the superposition principle

6. Write the formula for the electric field due to a point charge.

7. What is the direction of the electric field due to a positive point charge?

8. Explain why the electric field vector points away from a positive charge.

9. Compare the electric field at distances r and 2r from a point charge.

10. Why does the electric field decrease with distance squared?

11. Calculate the electric field at 0.5 m from a charge of 2μC.

12. A charge of +3μC produces an electric field of 10800N. Find the distance from the charge.

13. A proton is placed in an electric field of 5000N. Determine the force acting on it.

14. Two point charges and −2μC are placed 1 m apart. Determine the net field at the midpoint.

15. Explain how the direction of the electric field at a point depends on the signs of multiple charges nearby.

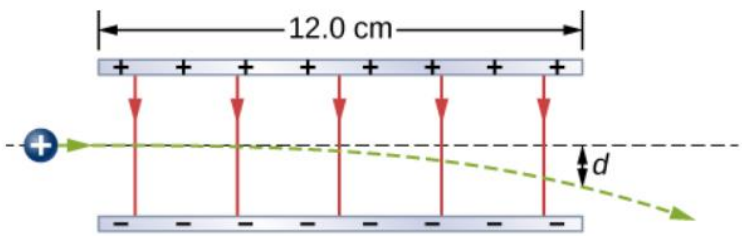
16. Compare the magnitudes of electric fields due to charges q_1 = 1μC and q_2 = 2μC at the same distance. Which one is more significant and why?

17. Two charges +2μC and −3μC are placed 1 m apart. Find the net electric field at a point 0.25 m.

18. Three charges q_1 = +1μC q_2 = +2μC, q_3=−1μC are at the vertices of a right triangle (1 m sides).

Find the net field at the right-angle vertex and its direction.

19. Create a diagram showing the electric field at several points around two unequal charges

20. A rod of length 1 m has a linear charge density λ=5μC/m. Calculate the electric field at a point 0.5 m from its midpoint (perpendicular bisector)

21. A proton enters the uniform electric field produced by the two charged plates shown below. The magnitude of the electric field is $4.0 \times 10^5 N/C$, and the speed of the proton when it enters is $1.5 \times 10^7 m/s$. What distance d has the proton been deflected downward when it leaves the plates?